Hyperspherical Structural-Aware Distillation Enhanced Spatial–Spectral Bidirectional Interaction Network for Hyperspectral Image Classification

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***Abstract*— The existing methods for hyperspectral image classification (HSIC) mainly focus on the extraction of spectral and spatial features while paying less attention to the interaction of each other. Besides, most of them directly use a parameterized classifier as the final layer of the network. While this design is convenient for end-to-end optimization with the backbone, it overlooks the utilization of the metric space. In this article, a novel hyperspherical structural-aware distillation enhanced spatial–spectral bidirectional interaction network (HSDBIN) is proposed for HSIC. HSDBIN uses a dual-branch design combining the 1-D CNN and transformer to separately learn the detailed spectral correlations and global spatial relationships in parallel. Then, by interacting and aggregating the independent information between two parallel branches, a bidirectional interaction block across branches is designed to explore complementary clues between spectral and spatial pipelines. Finally, to enhance the utilization of metric space and keep compact intraclass relationship, we propose a hyperspherical structural-aware distillation (HSD) to transfer the geometric relationship of hyperspherical space into the metric space of output logits. Extensive experiments and analysis on three public HSI datasets suggest the superiority of the proposed method and verify the effectiveness of the proposed modules.**

***Index Terms*— Bidirectional interaction, dual-branch, hyperspectral image, hyperspherical learning.**

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I. Introduction

HYPERSPECTRAL image classification (HSIC) is vital to many applications of remote sensing and has gained extensive attention [1], [2], [3]. In recent years, massive approaches based on machine learning (ML) have achieved surprisingly satisfactory performance in the HSIC task. These methods roughly fall into two types: traditional ML-based methods [4], [5] and deep learning (DL)-based methods [6], [7].

Early ML-based HSIC methods represented by SVM and its variants [8], [9], [10], sparse representation [11], and clustering algorithms [12], [13] mainly focus on the design and interaction between feature extractors and classifiers [14]. These traditional ML-based methods achieve competitive results, but they are prone to extract hand-crafted (or engineered) shallow features, which limits the performance gain of models.

With the rise of DL, models of diverse architectures specializing in HSIC tasks are designed [15], [16], [17], [18], [19], [20], and they are capable of tackling different issues [21], [22], [23], [24] and have achieved impressive performance on HSIC. CNN-based models, with their powerful capability to learn detailed and hierarchical representations, are predominantly developed in the HSIC framework and achieved remarkable results [25], [26], [27]. Different from the optical signal, hyperspectral data have their own inherent characteristics, spatially sparse while spectrally correlated, which is highly capable of providing a rich and detailed part-level description of objects [28], [29]. Therefore, the key factor for designing an excellent architecture for HSIC is to effectively extract spatial and spectral information in a collaborative manner [30], [31]. A straightforward solution would be to comprehensively extract the spectral–spatial features of the patch-level hyperspectral cube via the 3-D convolution kernels [32], [33], [34]. These 3-D-based models

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achieve impressive performance in HSIC, but they may suffer from heavy computation cost and the difficulties in balancing the contribution between the spatial and spectral pipelines. Another line of work is based on the dual-branch architecture [35], [36], [37], where the parallel design is adopted to learn spectral and spectral features separately and then fuse them by the concatenation operation or sum them based on a tradeoff parameter. The fashion of dual-branch networks makes it possible to flexibly design network structures tailored to data attributes of spectral and spatial pipelines. Broadly, some methods [36], [38] provided well-designed strategies for each branch, which effectively enhance the modeling capabilities of individual branches. However, learning the spectral and spatial features independently might be incompetent in capturing the complementary clues between spatial and spectral branches.

Furthermore, in addition to the careful design of the hyperspectral backbone, the selection of the linear classifier in the final layer is also crucial for overall model generalization. Leveraging the parametric Softmax classifier with cross-entropy loss as the classification layer has been the de facto choice for most DL-based HSIC methods [39], [40], [41]. This design allows the backbone model and the classifier to be jointly optimized by gradient descent algorithms in an end-to-end manner. However, some recent studies have shown a phenomenon called neural collapse [42], [43], [44], i.e., at the terminal phase of training, the within-class means of features and the classifier vectors converge to the vertices of a simplex equiangular tight frame (ETF). The classifier in the final layer of the model tends to converge to class centroids and exhibits the nearest class-center decision rule, i.e., the classifier's decision collapses to simply choosing whichever class has the closest train class mean. While this highly structured geometric relationship is convenient for training and optimization, it suffers from limited interclass separation and overly simplified intraclass representations. Moreover, using a simple linear classifier to fit the feature-to-semantic mapping function fails to fully use the metric space, leading to the concentration of interclass prototypes in partial regions of the overall metric space.

To bypass these limitations, some methods uniformly fix class prototypes throughout the entire hypersphere space and enforce equal similarity between any pairs of classes [45], [46]. This hypersphere-based metric learning approach not only enhances the utilization of the metric space but also ensures a larger margin of interclass separation. While this prior hyperspherical uniformity is theoretically elegant and ideal, the learning objective of uniformly filling the entire metric space with features from different classes is overly idealistic, making the model challenging to optimize. Alternatively, even if the model struggles to achieve the desired interclass distribution, severe distortions in the geometric structure of the data may cause the model to deviate from the true underlying predictive mechanisms. Therefore, integrating the highly structured geometric relationships of parameterized classifiers with the ideal interclass geometric distribution offered by hyperspherical prototype learning may serve as a more effective solution.

To address the aforementioned problems, a novel hyperspherical structural-aware distillation enhanced spatial–spectral bidirectional interaction network (HSDBIN) is proposed in this article. To model spectral correlations and global spatial relationships in parallel while capturing their complementary clues, we proposed a spatial–spectral bidirectional interaction network (SBINet). SBINet can learn detailed spectral features and global spatial relationships by the parallel 1-D CNN and transformer, and it integrates complementary spectral–spatial information between the two parallel branches by the bidirectional interactions across branches. To better integrate the complementary information across branches, we designed two interaction blocks tailored to each branch based on the inherent attributes of spatial and spectral features. Finally, to enhance the utilization of metric space and keep compact intraclass relationship, we propose a hyperspherical structural-aware distillation (HSD). This constraint aims to transfer the geometric structure of the hyperspherical space into the feature space of output logits. The main contributions of this article can be summarized as follows.

1) We introduced the fashion of bidirectional interaction into a hyperspectral image classification model and proposed an SBINet, where the parallel spectral–spatial dual-branch network is capable of independently learning detailed spectral correlations and global spatial relationships and the spectral–spatial bidirectional interaction block can capture the complementary clues between them.
2) Based on the specific attributes of spatial and spectral features, two interaction blocks tailored to each branch are designed. The spectral-to-spatial (Spe2Spa) interaction block can encode the spectral correlations to the spatial branch and dynamically focus on their relevant parts. The spatial-to-spectral (Spa2Spe) aggregation block is capable of fully uncovering underlying patterns within spatial features in a multiorder interaction manner and squeezing important information into the spectral pipeline.
3) We propose an HSD to enhance the utilization of metric space and preserve the highly structured relationship of output logits. HSD includes two pipelines, in which the hyperspherical prototype learning will enhance the utilization of metric space of output features and the structural-aware distillation can transfer the geometric structure from the hyperspherical space to the logits' metric space and maintain compact intraclass relationships.

The remainder of this article is organized as follows. In Section II, the proposed HSDBIN is presented in detail. Section III presents and analyzes the experimental results. Finally, conclusions are reported in Section IV.

II. Methodology

The overall framework of HSDBIN is illustrated in Fig. 1. First, a batch of labeled data and unlabeled data are sampled from the HSI, and they are a group of patch-level $p \times p$ hyperspectral cubes which include the center pixels and its

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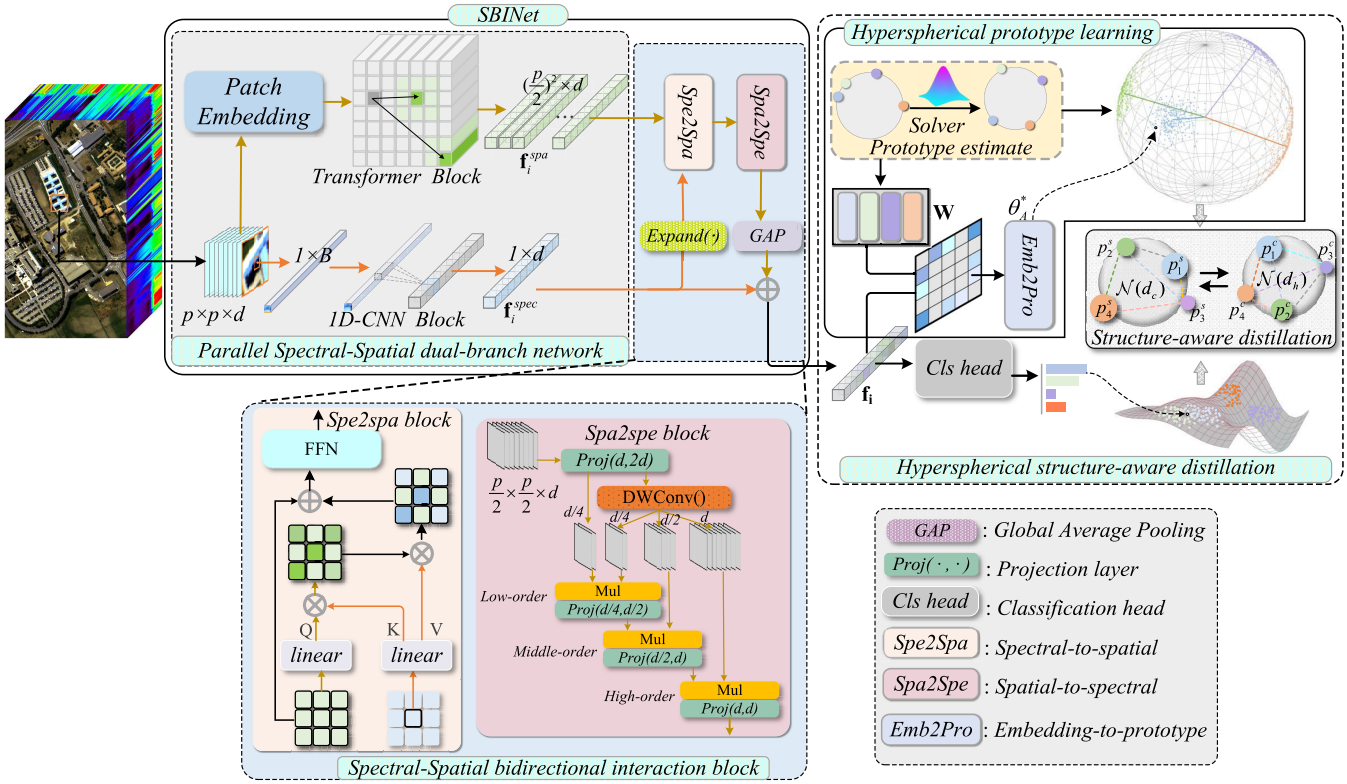


Fig. 1. Overview of the proposed HSDBIN, which is primarily composed of three key constituent modules, i.e., parallel spectral–spatial dual-branch network, SSBI block, and HSD.

surroundings. Then the parallel spectral–spatial dual-branch network is designed to independently learning detailed spectral correlations and global spatial relationships. Next, a spectral–spatial bidirectional interaction block is used to couple the complementary features between the two parallel branches. Finally, an HSD including hyperspherical prototype learning and structural-aware distillation is leveraged to constrain the optimization of SBINet.

A. Parallel Spectral–Spatial Dual-Branch Network

It is assumed that labeled samples $\mathbf{X}^s = \{\mathbf{x}_i^s\}_i^{n_s} \in \mathbb{R}^{1\times B}$ with the corresponding class labels $\mathbf{Y}^s = \{\mathbf{y}_i^s\}_i^{n_s} \in \mathbb{R}^{1\times C}$, where n_s and C denote the size of training samples and the number of class, respectively. $\mathbf{x}_i^{\text{spec}} \in \mathbb{R}^{1\times B}$ is the center pixel of samples, and we denote $\mathbf{x}_i^{spa} \in \mathbb{R}^{p\times p\times B}$ as the sampled hyperspectral patch based on the center pixel $\mathbf{x}_i^{\text{spec}}$. Typically, the central pixel incorporates the discriminative spectral feature, while the surroundings provide spatial context to condition the classification results. As shown in Fig. 2, motivated by the excellent modeling ability of 1-D CNN for 1-D signals, a 1-D CNN is used to learn the detailed spectral features. The 1-D CNN consists of a 1-D convolutional layer with 20 $1 \times k_0$ convolutional kernels, a batch normalization (BN) layer, a MaxPooling layer with $1 \times k_1$ kernel size, a projection layer, and a layer normalization (LN) layer. Following the paper [47], $k_0 = \text{ceil}(B/9)$ and $k_1 = \text{ceil}(k_0/5)$, then $\tanh(\cdot)$ is selected as the active function and is applied after MaxPooling and projection layers. It should be noted that the output of MaxPooing layer is flattened as a feature vector and the projection layer will map it as a d-dimensional vector to be consistent with the output dimension of the transformer network. A classical result in learning theory tells us that the parameters of the learner need to match the size of data, hence we only adopt one simple yet effective 1-D CNN block to learn latent spectral correlation to avoid the overleaning of spectral information. Then taking the labeled sample as an example, the output of the spectral branch can be expressed as

$$\mathbf{f}_i^{\text{spec}} = \theta_{\text{spec}}\left(\mathbf{x}_i^{\text{spec}}\right),\ \mathbf{f}_i^{\text{spec}} \in \mathbb{R}^{1\times d} \tag{1}$$

where θ_{spec} represents the overall 1-D CNN.

In terms of spatial branch, a vision transformer (ViT)-based network is introduced to extract the spatial features of hyperspectral patches. Following ViT, the hyperspectral patch is separated into nonoverlapping and smaller patches, and each patch is treated as a token. The transformer block is formed by a standard multihead self-attention and a feed-forward network (FFN) (an MLP layer and an LN layer), and the query dimension of each head is 48. To fully explore the discriminative spatial information, four transformer blocks are used to model long-range dependencies in the spatial branch. Denote $\theta_{\text{spa}}(\cdot)$ as the transformer network, then the output of the spatial branch can be written as

$$\mathbf{f}_i^{spa} = \theta_{\text{spa}}\left(\mathbf{x}_i^{spa}\right),\ \mathbf{f}_i^{spa} \in \mathbb{R}^{p/2\times p/2\times d} \tag{2}$$

where $\mathbf{f}_i^{spa}$ is the token map of the spatial hyperspectral cube.

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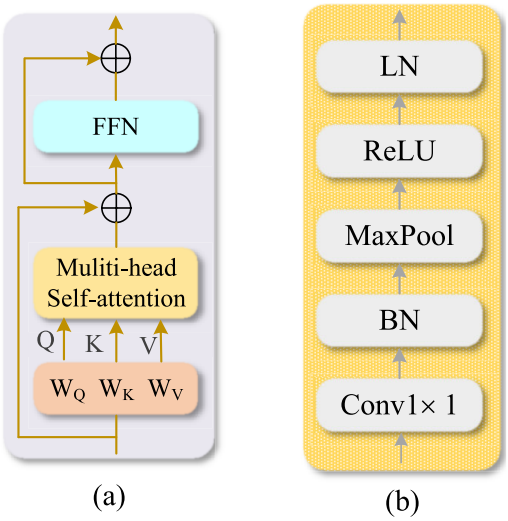


Fig. 2. Details of parallel dual-branch network. (a) Transformer block. (b) 1D-CNN block.

B. SSBI Block

To explore the complementary clues between spatial and spectral features, a bidirectional interaction module across branches is designed to integrate complementary information between the two parallel branches. While using conventional attention modules to directly aggregate features can also learn complementary patterns to some extent, this general structure often lacks the inductive biases specific to hyperspectral data, such as spatial sparsity and spectral continuity. For hyperspectral data, the spatial information typically includes correlations related to spatial positions. Therefore, in the Spe2Spa block, we primarily use cross-attention to encode central spectral information into the spatial branch. Conversely, the spectral branch usually contains more detailed information across different bands. Hence, in the Spa2Spe block, we use a multiorder interaction approach to thoroughly explore the interchannel correlations, thereby aggregating them into the spectral branch. Initially, for spectral interaction, a cross-attention block is used to encode the spectral information into the spatial pipeline. To mitigate the issue of imbalanced feature interaction ($1\times d$ versus $p^2/4\times d$), the expand operation is used to replicate and stack the central pixel to match the dimensions of the spatial tokens. The spectral feature vector is first expanded to the same spatial size as the spatial token map by the Expand$(\cdot)$ operation. Inspired by the design of [48], we apply the expanded spectral feature to the key and value in the spatial attention block like generic cross-attention. Mathematically, the output after spectral interaction can be expressed as

$$\begin{aligned} q &= \mathbf{f}_i^{\text{spa}} W_q, \quad k = \mathbf{f}_i^{\text{spec}} W_k, \quad v = \mathbf{f}_i^{\text{spec}} W_v \\ A &= \text{softmax}\left(qk^T/\sqrt{d/h}\right), \quad \mathbf{f}_i^{\text{SpeI}} = Av \end{aligned} \tag{3}$$

the $W_q, W_k, W_v \in \mathbb{R}^{d\times d}$ are the learnable weight matrices.

After encoding the spectral information into the spatial branch, a spatial interaction block is adopted to incorporate the coupled spectral–spatial into the spectral branch. In terms of the spatial interaction block, a multiorder interaction block is developed to fully explore the local and long-range relationship in the channel dimension of coupled features. The feature $\mathbf{f}_i^{\text{SpeI}}$ after the Spe2Spa interaction is first embedded into a set of projected features (multiorder features) $[\hat{\mathbf{f}}_i^{\text{SpeI},d_0}, \tilde{\mathbf{f}}_i^{\text{SpeI},d_0}, \ldots, \tilde{\mathbf{f}}_i^{\text{SpeI},d_{K-1}}] \in \mathbb{R}^{(p/2)^2\times d_k}$, where $d_k = d/2^{(K-k-1)}$ and K is the number of interacted orders. Notably, the first two terms of the multiorder interaction features are different features, which have the same dimensions but contain different components of the original features. Next, the multiorder interaction is performed in a recursively gated manner

$$\hat{\mathbf{f}}_i^{\text{SpeI},k+1} = \phi_k\left(\tilde{\mathbf{f}}_i^{\text{SpeI},k}\right) \odot \rho_k\left(\hat{\mathbf{f}}_i^{\text{SpeI},k}\right), \quad k = 0, 1, \ldots, K-1 \tag{4}$$

where $\{\phi_k(\cdot)\}$ are a set of depthwise convolution layers and $\{\rho_k(\cdot)\}$ are projection layers used to match the dimension in different orders. After carrying out the K-order interaction along the spectral dimension, the interacted features are aggregated into the spectral branch by a global average pooling (GAP) layer. The interaction order is a hyperparameter, and the Spa2spe block will degrade to a general self-attention module when $K = 1$, which cannot fully explore the interchannel correlations. Conversely, setting the interaction order too high may lead to overinteraction, resulting in a loss of long-range relationships. Therefore, we empirically set it to 3. Hence, the final output feature vector can be represented as

$$\mathbf{f}_i = \lambda \mathbf{f}_i^{\text{spec}} + (1-\lambda)\mathbf{f}_i^{\text{SpaI}}, \quad \mathbf{f}_i \in \mathbb{R}^d \tag{5}$$

where $\mathbf{f}_i^{\text{SpaI}} = \text{GAP}(\hat{\mathbf{f}}_i^{\text{SpeI},K-1})$ and λ is a tradeoff parameter. Notably, due to different feature transformations between spatial and spectral pipelines, the dual-branch architecture will inevitably introduce disparities in feature representation. However, the bidirectional interaction block will bridge this discrepancy by learning complementary information between the two types of features.

C. Hyperspherical Prototype Learning for HSIC

After obtaining the features extracted by the backbone network, the HSD first adopts a hyperspherical prototype learning to transform the feature into hyperspherical space. As known, most existing HSIC methods typically use a linear classifier to directly output the logits of each class. Subsequently, they use a cross-entropy loss to jointly optimize the classifier and backbone. Some studies yielded that the within-class means of features and the classifier vectors of such pipeline are prone to converge to the vertices of a simplex ETF at the terminal phase of training. This will lead to the classifier behavior collapsing into a nearest class-center decision criterion. Furthermore, this fully parameterized training scheme, due to ignoring known inductive biases (such as large margin separation, simplicity, and knowledge about source data), results in insufficient utilization of metric space [49]. Hyperspherical prototype learning can partially address these drawbacks, but it is challenging to optimize.

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To address this, a structure-aware distillation scheme is used to transfer the geometric structure of the hyperspherical metric space into the metric space of logits, guiding the learning and optimization of both the classifier and the backbone. This approach not only enhances the utilization of the metric space but also, thanks to the dual-classifier-like design, provides dual-perspective decision results for inference. First, we use a cost function based on a Gaussian latent kernel to estimate the universally optimal prototype configurations within the hypersphere. Denote $\mathcal{H}^d$ as the unit hypersphere of d-dimensional space and $\mathcal{P}_h(\cdot)$ as the prototypes' distribution over $\mathcal{H}^d$. The Gaussian potential kernel $G_t(\alpha,\beta) \triangleq \mathcal{H}^d \times \mathcal{H}^d \to \mathbb{R}_+$ can be represented as

$$G_t(\alpha,\beta) \triangleq e^{-t\|\alpha-\beta\|_2^2}, \quad t>0 \tag{6}$$

where t is a bandwidth parameter. Then the uniformity loss can be defined as

$$\mathcal{L}_{\text{pro}} \triangleq \log \mathop{\mathbb{E}}_{\mathrm{w}_i,\mathrm{w}_j \overset{\text{i.i.d.}}{\sim} \mathcal{P}_h} [G_t(\alpha,\beta)], \quad t>0 \tag{7}$$

where $\mathrm{w}_i, \mathrm{w}_j$ are the prototypes in the prototype matrix $\mathbf{W} \in \mathbb{R}^{d\times C}$ and $\mathbf{W}$ is randomly initialized in the first iteration. $\mathcal{L}_{\text{pro}}$ is nicely tied with the uniform distribution of points on $\mathcal{H}^d$. Notably, the optimization of $\mathcal{L}_{\text{pro}}$ is independent of the backbone and classifier. When the solution of $\mathbf{W}$ is obtained, a hyperspherical learning based on label-to-prototype assignment is leveraged to align features with their corresponding prototype. Let the mapping from the output feature to its assigned can be learned by

$$\mathcal{L}_{\text{Sph}} = \sum_{i=1}^{n_s} \frac{1}{2}\left(\mathbf{f}_i^\top \mathbf{W}\theta_A^*(\mathbf{y}_i^s) - 1\right)^2 \tag{8}$$

where $\theta_A^*(\cdot)$ is a bijective mapping from labels to prototypes. Here, we adopt a momentum update to find this mapping function, that is,

$$\theta_A^* = \arg\min_{\theta_A} \sum_{c=1}^{C}\left[-\bar{\mathbf{f}}_c^\top \mathbf{W}\theta_A(c) + \log\sum_{k=1}^{C} e^{\bar{\mathbf{f}}_k^\top \mathbf{W}\theta_A(k)}\right] \tag{9}$$

where $\bar{\mathbf{f}}_c$ is the momentum averaging over the representations of the cth class and the second term is constant for all the elements of θ_A^*.

D. Structure-Aware Distillation and Overall Optimization

To transfer the geometric structure of the hyperspherical metric space into the metric space of logits, a structure-aware distillation scheme is established for HSD. Specifically, through the optimization of (8), features are pushed toward their corresponding prototypes. Afterward, we aim to preserve this structural information in the output logits of the classifier. First, we define pairwise distances in two metric spaces as

$$\begin{aligned} d_c(\mathbf{f}_i,\mathbf{f}_j) &= \left\|\rho_c(\mathbf{f}_i) - \rho_c(\mathbf{f}_j)\right\| \\ d_h(\mathbf{f}_i,\mathbf{f}_j) &= \left\|\rho_h(\mathbf{f}_i) - \rho_h(\mathbf{f}_j)\right\| \end{aligned} \tag{10}$$

where $\rho_c(\cdot) : \mathbb{R}^d \to \mathbb{R}^C$ is the classification head and $\rho_h(\mathbf{f}_i) = \mathbf{f}_i^\top \mathbf{W}$ is the hyperspherical mapping. Subsequently, to measure the structural similarity between two different dimensions of hypersphere spaces, the normal distribution [50] is introduced to reconstruct the pairwise distances, i.e.,

$$\begin{aligned} \mathcal{N}(d_c) &= N_c \frac{1}{\sigma_c\sqrt{2\pi}} \exp\left[-\frac{1}{2}\frac{(d_c-\mu_c)^2}{\sigma_c^2}\right] \\ \mathcal{N}(d_h) &= N_h \frac{1}{\sigma_h\sqrt{2\pi}} \exp\left[-\frac{1}{2}\frac{(d_h-\mu_h)^2}{\sigma_h^2}\right] \end{aligned} \tag{11}$$

where N_c, N_h are constants for data normalization. Next, the objective of structure-aware distillation is defined as a formal of the binary cross-entropy

$$\mathcal{L}_{\text{Dis}} = \mathcal{N}(d_c)\log\mathcal{N}(d_h) - (1-\mathcal{N}(d_c))\log(1-\mathcal{N}(d_h)). \tag{12}$$

Notably, denote the batch size as n_b, d_c and d_h are THE matrices of pairwise distance and their dimension is $n_b \times n_b$. Therefore, $\mathcal{L}_{\text{Dis}}$ enables knowledge distillation across different feature dimensions.

Denoted the cross-entropy loss of classifier as $\mathcal{L}_{cls}$ and it can be calculated by

$$\mathcal{L}_{\text{Cls}} = \frac{1}{n_s}\sum_{i=1}^{n_s}\sum_{c=1}^{C} -y_i^c \log p_i^c \tag{13}$$

where $p_i = \rho_c(\mathbf{f}_i)$ is the probabilistic output of classification head. Let $\theta(\cdot)$ be the overall features' mapping of SBINet, and the overall entire objective function can be represented as

$$\mathcal{L} = \mathcal{L}_{\text{Cls}} + \lambda_1\mathcal{L}_{\text{Sph}} + \lambda_2\mathcal{L}_{\text{Dis}}. \tag{14}$$

Although fixed prototypes are slower to converge compared with dynamic prototypes, they can learn more accurate prototypes based on predefined prior information. While dynamic prototype estimation can accelerate model convergence, the absence of fixed prototype points, as in hyperspherical prototypes, makes it difficult to fully use the metric space. Moreover, we just use it as an auxiliary loss, and the classification loss can accelerate its convergence.

III. Experimental Results and Analysis

In this section, three benchmark HSI datasets with disjoint training and testing data are introduced to verify the effectiveness of our method. Furthermore, the results of the proposed method and several relevant algorithms methods are compared. Then, ablation experiments are designed to explore the effectiveness of the proposed constituent modules, and some visual maps are presented. Finally, some model-specific parameters are discussed and analyzed.

A. Datasets Description

1) Houston2013: The Houston2013 dataset comprises 349 × 1905 pixels with a spatial resolution of 2.5 m and 144 spectral channels in a range from 0.38 to 1.05 μm, which was collected by the Compact Airborne Spectrographic Imager (CASI) sensor. Besides, the training and testing sets of this dataset are provided by the Image Analysis and Data Fusion

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TABLE I
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE HOUSTON2013 DATASET

No.	Color	Land cover type	Training Set	Testing Set
1		Healthy Grass	198	1053
2		Stressed Grass	190	1064
3		Synthetic Grass	192	505
4		Tree	188	1056
5		Soil	186	1056
6		Water	182	143
7		Residential	196	1072
8		Commercial	191	1053
9		Road	193	1059
10		Highway	191	1036
11		Railway	181	1054
12		Parking Lot1	192	1041
13		Parking Lot2	184	285
14		Tennis Court	181	247
15		Running Track	187	473
Total		-	2832	12197

TABLE II
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE PAVIA UNIVERSITY DATASET

No.	Color	Land cover type	Training Set	Testing Set
1		Asphalt	548	6304
2		Meadows	540	18146
3		Gravel	392	1815
4		Trees	524	2912
5		Metal Sheets	265	1113
6		Bare Soil	532	4572
7		Bitumen	375	981
8		Bricks	514	3364
9		Shadows	231	795
Total		-	3921	40002

Technical Committee of the IEEE Geoscience and Remote Sensing Society (GRSS) in the 2013 Data Fusion Contest.[1] The details of training and testing sets are listed in Table I. The false-color image and distribution of samples are presented in Fig. 3.

2) Pavia University: The Pavia University (PaviaU) dataset[2] covers the surrounding urban area of Pavia city and is gathered by the Reflective Optics System Imaging Spectrometer (ROSIS), which has 1.3-m spatial resolution and the wavelength scope of 0.43–0.86 μm. It contains 610 $\times$ 340 pixels and 103 bands after pruning the noisy channels. The class-specific training and testing sizes and the corresponding land-cover types for PaviaU datasets are reported in Table II. The false-color image and the distributions of training and testing samples are shown in Fig. 4.

[1]http://www.classic.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

[2]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

TABLE III
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE YANCHENG DATASET

No.	Color	Land cover type	Training Set	Testing Set
1		Sea	209	2186
2		Offshore area	140	1448
3		Salt field	6	104
4		Pond	20	173
5		Partina anglica	7	76
6		Mudflats	25	243
7		Aquaculture pond	25	238
8		Paddy filed	87	745
9		Estuarine area	27	248
10		River	27	272
11		Woodland	19	196
12		Barren	25	129
13		Building	37	489
14		Fallow land	26	208
15		Rainfed cropland	28	176
16		Suaeda salsa	14	71
17		Irrigation canal	10	28
18		Phragmites	12	120
Total		-	744	7150

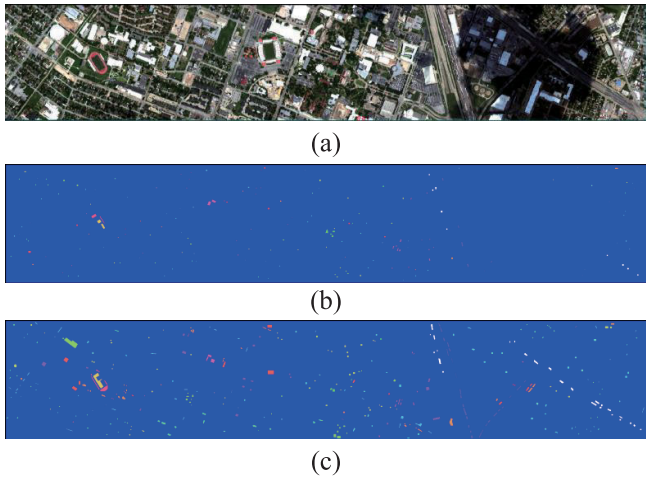


Fig. 3. Houston2013 dataset. (a) False-color image. (b) Training set. (c) Testing set.

3) YanCheng Wetlands: The YanCheng wetlands (YC) dataset [51] was captured by the ZY1-02D AHSI sensor on April 4, 2019. As shown in Fig. 5, the spatial size of the YC dataset is 1175 $\times$ 585 pixels with a spatial resolution of 30 m and mainly covers the coastal area of Yancheng, Jiangsu, China. Yancheng is rich in wetland resources, with a large area of radiating sandbars, coastal forest farms, and grasslands. After removing bad bands, it contains 147 bands from the spectral range of 0.4–2.5 μm. The detailed split information of training and testing samples are summarized in Table III.

B. Implementation Details

The HSDBIN algorithm is implemented with Python and the PyTorch framework, and all the experiments are implemented on the same hardware platform GPU: GTX-3090, CPU:

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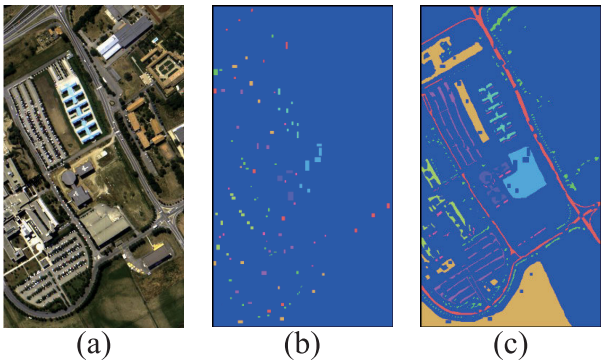


Fig. 4. PaviaU dataset. (a) False-color image. (b) Training set. (c) Testing set.

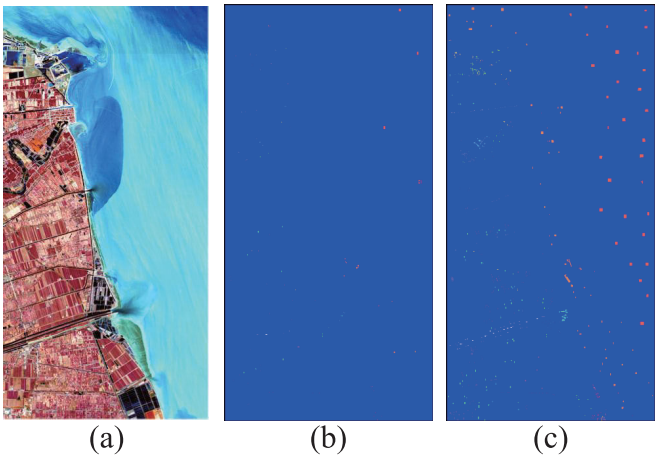


Fig. 5. YC dataset. (a) False-color image. (b) Training set. (c) Testing set.

Intel 4210R, and memory: 90G. For all the experiments, the spatial size and feature dimension of transformer's token are set as 2×2 and 192, respectively. The number of heads of the self-attention block in transformer and cross-attention block in the Spe2Spa module is set as 4. The Adam is adopted as the optimizer with $1e^{-3}$ learning rate, and the weight decay is set to $1e^{-4}$. The training epochs are 100, and we randomly select 10% samples of the training data to serve as the validation set for developing the models. It is important to note that our algorithm's optimization consists of two independent pipelines. First, the fixed prototype points in the high-dimensional hypersphere space (i.e., 192 dimensions) are obtained by optimizing (7). Then, based on these hyperspherical prototypes, the optimal model parameters are obtained by optimizing (14). The overall accuracy (OA), class-specific accuracy, average accuracy (AA) and the Kappa coefficient are used to evaluate the classification performance. To avoid the biased estimate, all the experiments were conducted with ten independent tests.

C. Comparison Results and Analysis

To verify the classification performance of HSDBIN, eight advanced HSIC methods with different backbone and learning strategy are selected for comparison, including SVM, A-SPN [26], 1-D + 2-D, which used a 2-D network to replace the Transformer of SBINet while keeping other structures in place, SSRN [35], DBDA [37], MAEST [39], [52], SGDCN [40], and SSFTT [41]. All the methods are implemented in the same hardware platform and the hyperparameters of all the counterparts are consistent with the optimal configurations in original papers. It is noteworthy that the separate datasets used for evaluation are more challenging compared with random sampling. Because the training and test data are disjoint and do not overlap with each other. In contrast, random sampling may lead to the implicit use of the features from the testing samples when training and testing samples are close to each other, and it may lead the model to learn a trivial solution by taking the shortcut.

The quantitative results of different methods on three datasets are reported in Tables IV–VI. Compared with other methods, the proposed HSDBIN achieves the best performance in terms of the OA, class-specific accuracy, and Kappa coefficient. Obviously, there are significant performance margins between HSDBIN and its counterparts, i.e., for the metrics of OA, AA, and Kappa, it surmounts the suboptimal values 2.95%, 3.15%, and 0.03 on the Houston2013 dataset, 5.19%, 5.77%, and 0.07 on the PaviaU dataset, and 2.19%, 2.24%, and 0.02 on the YC dataset. As is well-known, the Houston dataset poses challenges for HSIC methods, primarily due to the existence of distribution shifts between test and training samples. However, the HSDBIN still ranks the first place up to four categories Houston datasets, especially for the tenth class, "Highway," the HSDBIN outperforms other methods by a substantial margin (exceeding 34.15% over the second-based A-SPN). These observations yield that our method is not only effective in extracting spatial–spectral features and using metric space efficiently but also exhibits a certain level of robustness to the data with distribution shifts. It can be inferred that the HSDBIN is capable of learning the underlying and discriminative information even when the distribution of empirically observed samples slightly differs (discontinuous) from the expected distribution.

For the Pavia dataset, there is some imbalance in the distribution of different classes, with a significant difference observed between the second and ninth classes in the test set, where the ninth class has ten times fewer samples than the second class. Although the training samples are relatively balanced, this poses challenges for effectively using the metric space. Clearly, our method performs well in classifying each category, suggesting the superiority of the proposed approach.

The YC dataset, which has the largest number of categories, presents significant classification challenges. From the SVM classification results, it is evident that some categories are not linearly separable. Therefore, it is crucial for the network to effectively extract discriminative features. In addition, with its numerous categories and highly imbalanced sample distribution, this dataset poses significant difficulties. Surprisingly, our method not only achieved the highest OA but also performed well in classifying each individual category. This

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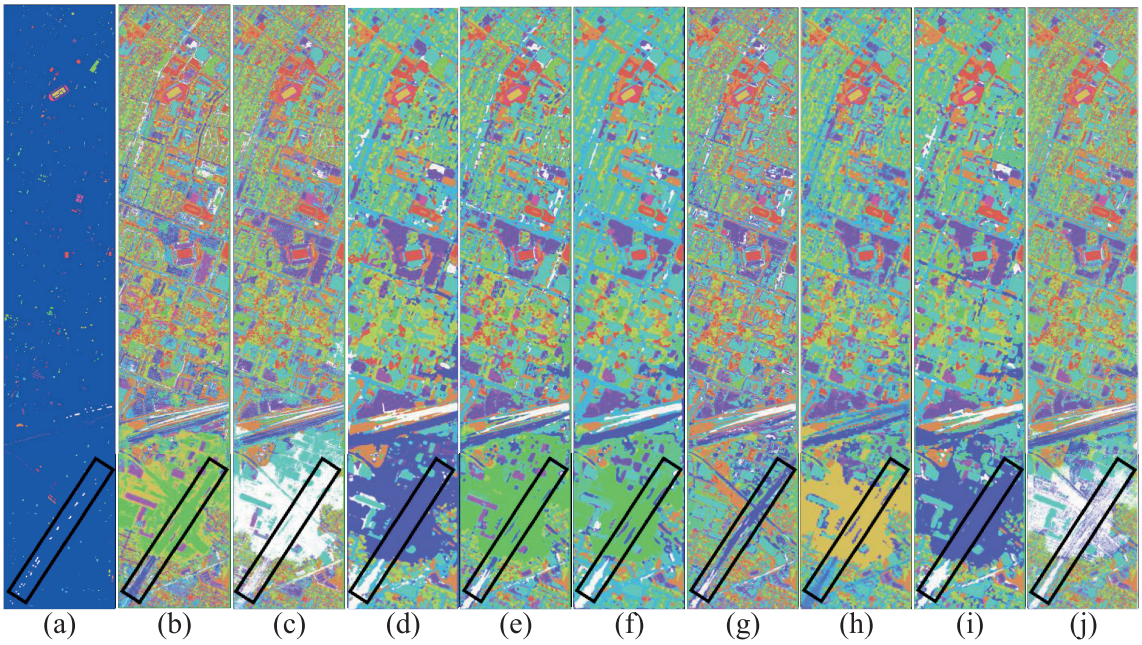


Fig. 6. Classification maps of different methods on the Houston2013 dataset. (a) Ground truth. (b) SVM. (c) 1-D + 2-D. (d) A-SPN. (e) SSRN. (f) DBDA. (g) MAEST. (h) SGDCN. (i) SSFTT. (j) HSDBIN.

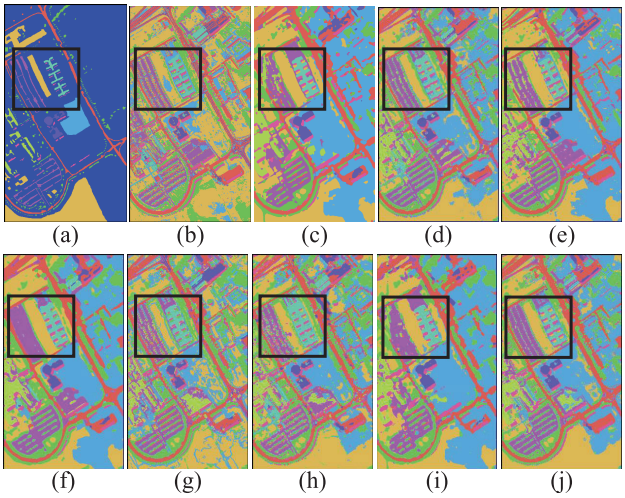


Fig. 7. Classification maps of different methods on the PaviaU dataset. (a) Ground truth. (b) SVM. (c) 1-D + 2-D. (d) A-SPN. (e) SSRN. (f) DBDA. (g) MAEST. (h) SGDCN. (i) SSFTT. (j) HSDBIN.

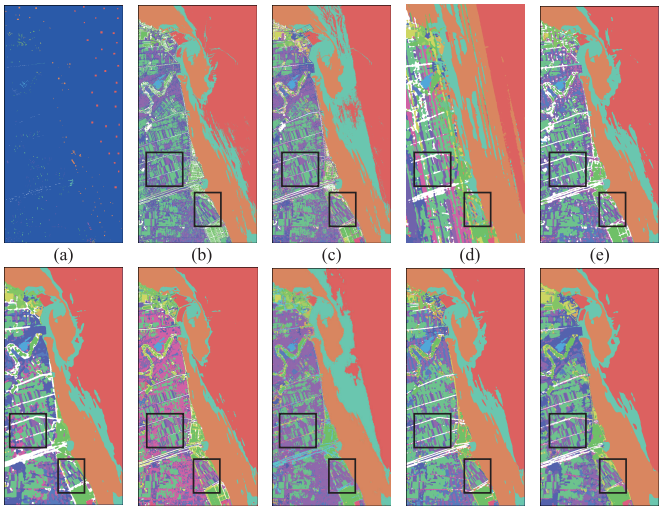


Fig. 8. Classification maps of different methods on the YC dataset. (a) Ground truth. (b) SVM. (c) 1-D + 2-D. (d) A-SPN. (e) SSRN. (f) DBDA. (g) MAEST. (h) SGDCN. (i) SSFTT. (j) HSDBIN.

improvement can be largely attributed to our bidirectional interaction network's ability to extract discriminative and complementary spatial–spectral features. Our HSD effectively enhances the utilization of the metric space to counteract this long-tail effect.

The qualitative classification maps are presented in Figs. 6–8, which suggest that the results of HSDBIN have clearer classification boundaries. In the black rectangular area, due to the lack of training data and the cover of cloud, pixels in the cloud-covered area and other areas have different spectral characteristics, resulting in the phenomenon that many remarkable methods can only make chance-level predictions. Remarkably, the HSDBIN is still capable of well-classifying the highway class while keeping relatively clear boundaries. Not surprisingly, HSDBIN exhibits the best classification map on the PaviaU dataset, especially in the area of black rectangular, HSDBIN can obtain clearer and smoother results than other counterparts. The YC dataset comprises numerous categories, and the land-cover distributions of coastal area are complex. However, it can be observed from the area of black rectangular that our method still achieved satisfactory classification maps on this dataset.

These observations imply that HSDBIN can extract more effective information and essential properties of a pixel to facilitate the networks to make more accurate decisions. Moreover, HSD can enhance the utilization of the metric space, thereby driving features to have larger interclass separation to combat the long-tail effect and potential out-of-

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TABLE IV
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

Class	SVM [8]	1D+2D	A-SPN [26]	SSRN [35]	DBDA [37]	MAEST [39]	SGDCN [40]	SSFTT [41]	HSDBIN
1	81.86±0.00	82.17±0.27	81.71±0.13	82.39±0.41	82.92±0.18	**84.50±0.72**	81.93±0.16	82.62±0.25	81.71±0.45
2	84.12±0.00	83.39±0.33	83.98±0.23	84.47±0.76	84.76±0.48	**99.18±0.40**	83.69±1.01	83.18±0.49	84.53±0.05
3	99.01±0.00	99.95±0.09	100.00±0.00	99.33±1.06	99.88±0.16	88.20±4.48	**100.00±0.00**	97.58±2.68	**100.00±0.00**
4	91.00±0.00	89.68±0.23	91.88±0.88	93.16±0.28	92.39±3.08	97.42±0.96	91.29±1.28	**97.33±3.16**	92.75±1.29
5	96.59±0.00	99.01±1.34	**100.00±0.00**	99.53±0.95	99.94±0.08	99.41±0.44	97.56±2.55	99.77±0.18	99.94±0.11
6	91.61±0.00	98.08±1.34	100.00±0.00	96.64±1.68	96.50±1.40	93.01±2.21	99.48±0.30	92.30±4.13	99.86±0.28
7	76.59±0.00	72.27±2.15	79.48±0.65	85.71±0.22	**91.16±2.87**	88.97±4.70	73.14±5.69	86.64±2.90	89.52±2.33
8	24.31±0.00	62.49±8.23	**78.01±0.15**	59.83±6.81	70.22±9.11	74.49±3.63	71.44±2.27	68.11±0.85	76.35±1.30
9	79.04±0.00	72.97±6.47	86.25±1.83	78.85±1.36	80.59±9.14	76.28±1.88	82.32±9.25	80.43±2.41	**88.22±2.49**
10	26.06±0.00	68.03±12.01	58.76±3.13	47.24±2.31	49.94±6.62	55.38±8.60	51.83±4.95	44.46±3.84	**92.93±6.03**
11	64.23±0.00	82.66±4.13	**99.66±0.30**	79.64±1.75	74.84±1.80	83.68±6.05	61.34±11.32	75.92±1.83	86.66±5.53
12	35.16±0.00	78.77±5.69	**98.14±1.02**	90.89±3.08	85.15±10.20	76.52±4.24	90.23±4.53	84.21±3.53	95.35±0.76
13	28.77±0.00	79.74±1.81	71.44±4.87	85.82±5.24	87.51±2.13	69.65±6.15	86.93±3.76	**90.88±2.82**	89.97±1.51
14	97.98±0.00	**100.00±0.00**	100.00±0.00	**100.00±0.00**	100.00±0.00	96.97±1.91	99.90±0.18	95.22±1.50	**100.00±0.00**
15	98.10±0.00	98.73±0.03	**100.00±0.00**	99.62±0.36	99.32±1.06	93.18±2.84	97.89±1.27	93.45±4.40	98.73±0.00
OA(%)	68.74±0.00	81.44±0.91	87.07±0.32	82.50±0.43	83.44±0.82	84.38±0.82	81.03±0.93	82.25±0.49	**90.02±0.50**
AA(%)	71.63±0.00	84.53±0.69	88.62±0.37	85.54±0.26	86.34±0.61	85.29±0.91	84.79±0.95	84.81±0.60	**91.77±0.37**
Kappa	0.66±0.00	0.80±0.01	0.86±0.00	0.81±0.00	0.83±0.00	0.83±0.10	0.80±0.01	0.81±0.00	**0.89±0.01**

TABLE V
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE PAVIAU DATASET

Class	SVM [8]	1D+2D	A-SPN [26]	SSRN [35]	DBDA [37]	MAEST [39]	SGDCN [40]	SSFTT [41]	HSDBIN
1	70.11±0.00	85.03±1.51	91.91±1.69	93.27±0.45	92.08±1.60	80.36±3.93	88.34±8.98	81.34±3.46	**94.83±1.64**
2	73.98±0.00	86.37±5.95	72.51±0.58	73.79±4.42	72.34±1.89	95.54±1.88	94.77±2.44	93.72±6.97	**95.91±0.71**
3	30.03±0.00	65.38±6.31	86.26±0.97	73.80±6.52	85.61±4.44	70.46±3.68	74.19±14.64	67.78±4.31	**88.34±4.65**
4	98.50±0.00	**99.57±0.14**	91.66±0.89	96.68±3.49	96.02±0.65	96.75±1.19	98.04±1.00	96.50±0.77	99.05±0.21
5	99.37±0.00	99.96±0.04	100.00±0.00	99.82±0.15	99.73±0.08	99.53±0.28	99.12±0.79	98.85±0.67	**100.00±0.00**
6	35.28±0.00	87.98±7.07	99.30±0.07	98.84±1.29	**99.71±0.17**	78.76±6.57	77.55±22.39	87.22±11.90	98.68±2.41
7	89.30±0.00	84.93±5.50	98.06±0.21	97.88±1.07	**99.20±1.49**	83.67±2.89	89.99±5.41	84.22±1.33	96.31±2.87
8	93.25±0.00	97.95±0.84	88.65±0.90	98.60±1.40	98.37±0.98	96.76±0.87	95.55±1.41	96.02±1.84	**99.79±0.05**
9	**99.87±0.00**	97.36±0.81	99.85±0.05	99.25±0.441	98.59±0.42	91.87±0.84	99.32±0.43	94.94±1.47	98.84±0.38
OA(%)	71.97±0.00	83.94±0.22	87.89±2.79	85.31±1.93	85.04±1.18	90.03±1.11	91.25±2.42	90.18±2.40	**96.44±0.29**
AA(%)	76.65±0.00	92.02±0.20	89.39±1.81	92.45±0.54	93.52±0.72	88.16±1.08	91.06±1.77	88.95±1.41	**96.83±0.61**
Kappa	0.63±0.00	0.79±0.00	0.84±0.03	0.81±0.02	0.81±0.01	0.80±0.00	0.88±0.03	0.87±0.03	**0.95±0.00**

TABLE VI
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE YC DATASET

Class	SVM [8]	1D+2D	A-SPN [26]	SSRN [35]	DBDA [37]	MAEST [39]	SGDCN [40]	SSFTT [41]	HSDBIN
1	97.99±0.00	92.92±1.65	80.17±0.65	88.77±5.92	99.58±0.46	85.87±7.69	97.99±2.61	99.07±1.00	**99.86±0.09**
2	100.00±0.00	100.00±0.00	91.82±0.16	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
3	2.88±0.00	25.19±3.57	**73.08±0.00**	31.73±21.73	17.88±4.20	68.23±8.19	56.01±17.37	5.77±5.12	66.73±4.45
4	0.00±0.00	75.84±2.79	84.16±0.46	89.94±6.15	75.49±1.70	81.16±2.90	79.05±7.37	56.65±17.48	**95.72±1.62**
5	0.00±0.00	100.00±0.00	99.21±1.05	78.16±30.62	**100.00±0.00**	91.38±3.23	98.68±1.61	39.47±30.61	40.26±15.56
6	92.59±0.00	**94.16±0.60**	92.43±0.20	91.93±1.92	94.07±1.24	80.16±3.68	86.11±2.48	89.05±7.89	93.99±0.20
7	**100.00±0.00**	82.61±5.68	90.08±0.57	95.04±2.89	99.50±0.81	89.69±3.21	96.95±1.16	98.57±0.98	94.96±4.69
8	**100.00±0.00**	99.76±0.42	86.63±0.25	99.33±0.54	98.50±0.68	98.62±0.61	98.12±1.95	99.84±0.20	99.57±0.05
9	96.77±0.00	96.77±0.77	97.98±0.36	**99.27±0.69**	94.92±0.94	96.73±4.15	95.56±2.60	95.97±3.40	96.77±0.62
10	29.04±0.00	90.38±5.06	86.10±0.54	88.60±10.81	98.68±0.64	87.45±1.15	86.40±1.35	98.53±0.62	**99.12±1.03**
11	100.00±0.00	100.00±0.00	100.00±0.00	99.90±0.20	100.00±0.00	99.69±0.77	78.70±33.43	100.00±0.00	**100.00±.00**
12	96.77±0.00	100.00±0.00	100.00±0.00	98.91±1.44	98.76±0.62	92.42±2.10	95.93±7.05	92.25±0.85	**100.00±0.00**
13	29.04±0.00	92.84±1.49	94.56±0.48	93.82±2.19	95.05±1.92	65.03±5.24	79.14±10.35	89.69±11.06	**98.69±0.28**
14	**100.00±0.00**	97.50±2.74	86.06±0.30	85.48±4.78	83.85±4.01	92.39±3.51	63.82±23.91	83.27±4.77	96.73±1.15
15	**100.00±0.00**	98.18±1.10	53.98±2.33	86.48±2.63	83.18±6.67	83.66±3.25	89.49±5.12	90.57±4.65	95.68±2.48
16	51.74±0.00	84.39±1.38	69.01±0.00	72.96±3.81	67.89±5.59	67.30±9.56	70.42±2.63	63.94±2.11	**84.66±5.24**
17	87.50±0.00	86.43±2.67	71.43±0.00	74.26±23.84	6.43±4.74	85.00±3.93	68.75±12.21	80.00±16.38	**95.00±4.84**
18	**92.05±0.00**	59.17±1.49	54.17±0.00	58.33±5.11	57.50±0.00	48.00±3.32	67.91±19.06	53.33±4.77	60.00±3.25
OA(%)	84.73±0.00	93.38±0.53	85.82±0.27	91.86±2.43	94.84±0.21	90.74±3.08	92.43±3.74	93.24±1.50	**97.03±0.32**
AA(%)	59.86±0.00	87.58±0.50	83.94±0.31	85.16±2.89	81.74±0.40	84.35±1.20	83.84±3.28	79.78±3.54	**89.82±1.29**
Kappa	0.82±0.00	0.92±0.01	0.83±0.00	0.90±0.03	0.94±0.00	0.89±0.04	0.91±0.04	0.92±0.02	**0.96±0.00**

distribution samples. Overall, compared with some advanced methods, quantitative and qualitative experiments on two datasets convey that HSDBIN not only delivers the promising classification accuracy but also gives clearer classification maps.

D. Ablation Study and Visualization

1) Ablation Studies of HSDBIN: To further explore the effectiveness of different components in SSBI-SA, ablation results with different constituent modules are reported in Table VII, where OA, AA, and kappa are used as mainly evaluation metrics. It can be obviously seen that the spectral-only (spectral branch) and spatial-only (spatial branch) models have substantial gaps with the two-branch design. After performing the spectral–spatial bidirectional interaction (SSBI), the model yields significant performance gains, indicating the remarkable effectiveness of the bidirectional interaction module. The bidirectional interaction block will capture the complementary cues between spectral and spatial branches

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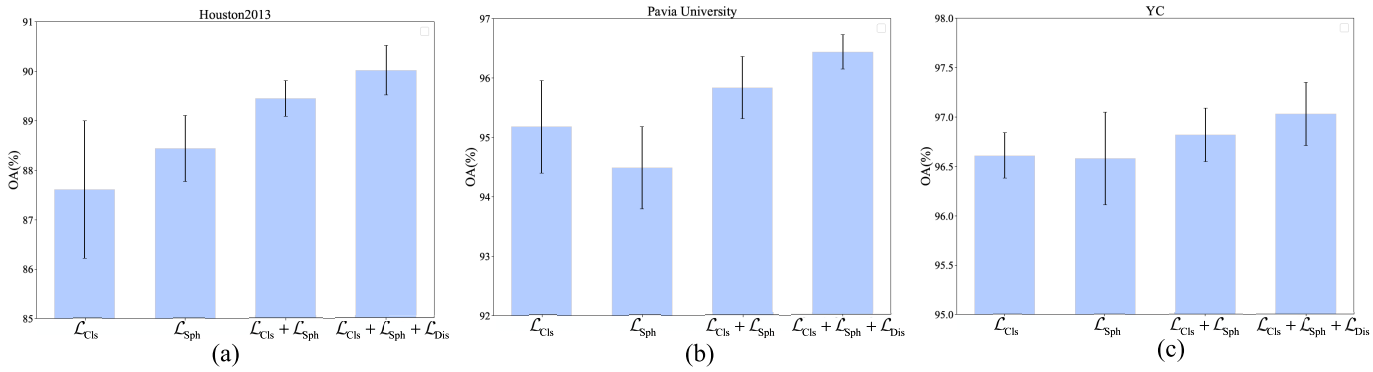


Fig. 9. OAs of HSDBIN with different learning schemes on three datasets. (a) Houston2013. (b) PaviaU. (c) YC.

TABLE VII
ABLATION STUDY OF HSDBIN ON THREE DATASETS

Datasets	Modules				Results		
	Spectral branch	Spatial branch	SSBI	HSD	OA	AA	Kappa
Houston2013	✓	×	×	×	81.82±0.76	84.66±0.57	0.80±0.01
	×	✓	×	×	86.04±0.60	88.66±0.43	0.85±0.01
	✓	✓	✓	×	87.61±1.39	89.95±1.03	0.87±0.01
	✓	✓	✓	✓	**90.02±0.50**	**91.77±0.37**	**0.89±0.01**
PaviaU	✓	×	×	×	79.64±0.87	86.08±0.41	0.71±0.01
	×	✓	×	×	93.60±0.62	93.68±0.49	0.91±0.01
	✓	✓	✓	×	95.18±0.78	95.45±1.13	0.94±0.01
	✓	✓	✓	✓	**96.44±0.29**	**96.83±0.61**	**0.95±0.00**
YC	✓	×	×	×	95.39±0.38	89.54±0.63	0.95±0.00
	×	✓	×	×	96.07±0.06	87.96±0.43	0.95±0.00
	✓	✓	✓	×	96.61±0.23	88.24±1.26	0.96±0.00
	✓	✓	✓	✓	**97.03±0.32**	**89.82±1.29**	**0.96±0.00**

to provide more diverse and discriminative components for HSIC task.

Obviously, when the HSD (i.e., hyperspherical prototype learning $\mathcal{L}_{\text{Sph}}$ and structural-aware distillation $\mathcal{L}_{\text{Dis}}$) is incorporated into backward optimization of the bidirectional interaction network, there is a significant improvement in model performance (with OA metrics increasing by 2.41%↑, 1.26%↑, and 0.42%↑ on the Houston2013, PaviaU, and YC datasets, respectively). This indicates that our proposed HSD enhances the utilization of metric space and helps the model make decisions closer to the real prediction mechanism. As expected, the HSDBIN exhibits the best performance when all the modules are added, verifying the reasonable solution that each partition of HSDBIN is productive and efficient for achieving better classification.

2) Effectiveness of the HSD: To further explore the effectiveness of HSD, we designed a more comprehensive ablation study to observe the performance improvement brought by each pipeline. Specifically, while keeping other components constant, we designed four variants: using only the classifier ($\mathcal{L}_{\text{Cls}}$), using only hyperspherical prototype learning ($\mathcal{L}_{\text{Sph}}$), combining the classifier with hyperspherical prototype learning ($\mathcal{L}_{\text{Cls}} + \mathcal{L}_{\text{Sph}}$), and combining the classifier and hyperspherical prototype learning with structural-aware distillation constraints ($\mathcal{L}_{\text{Cls}} + \mathcal{L}_{\text{Sph}} + \mathcal{L}_{\text{Dis}}$). These variants' OAs across the three datasets are shown in Fig. 9. It can be observed that our method achieves the best classification results after incorporating all the three losses. In our method, our hyperspherical structural distillation includes both hyperspherical prototype learning and structural-aware distillation. While hyperspherical prototype learning is a special form of prototype learning, hyperspherical loss can also optimize classification tasks. Interestingly, $\mathcal{L}_{\text{Cls}}$-only or $\mathcal{L}_{\text{Sph}}$-only schemes does not show a clear advantage over each other individually. However, when both $\mathcal{L}_{\text{Cls}}$ and $\mathcal{L}_{\text{Sph}}$ are optimized simultaneously for the backbone, the model's performance shows a significant improvement. This is because $\mathcal{L}_{\text{Cls}}$ and $\mathcal{L}_{\text{Sph}}$ contribute to enhancing the semantic information of features and the utilization of the metric space. However, without structural-aware distillation, the logits output by all the classifiers cannot retain the geometric structure information of the features in the latent space. As expected, the model's performance significantly improved after adding $\mathcal{L}_{\text{Dis}}$, validating the effectiveness of our approach. Using structural-aware distillation to transfer the geometric information from the hyperspherical space to the metric space of classifier outputs, the utilization of the metric space will be enhanced (resulting in greater interclass separation and tighter intraclass distances) and the optimization of the model will be accelerated.

3) Visualization: To intuitively understand the effectiveness of different submodules, Fig. 10 visualizes the t-SNE maps of the original HSI data, projection of HSDBIN, and projection with HSD. To visualize the utilization of the metric space,

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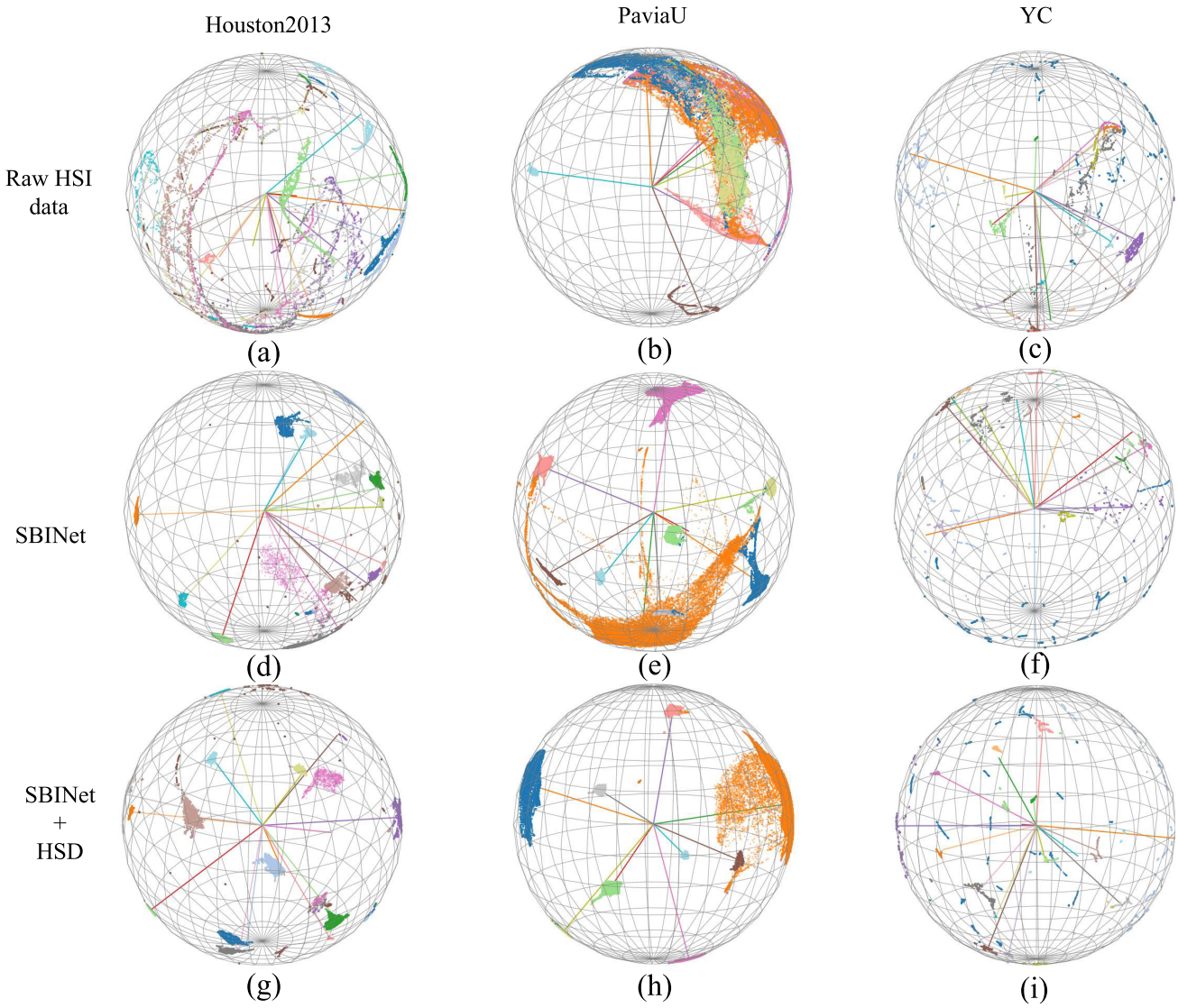


Fig. 10. Projection of different data points on three dataset. The scatter points with different colors on spheres are the feature embeddings of different class, and the intersections of light dash-dot lines and sphere are the class features (mean vectors). Each rows represent different projections, the first row is the raw HSI data, the second row is the outputs of SBINet, and the third row presents the results after incorporating HSD enhancement. The first column depicts results for the Houston dataset, the second column for the Pavia dataset, and the third column for the YC dataset. (a) Raw data of Houston2013 dataset. (b) Raw data of PaviaU dataset. (c) Raw data of YC dataset. (d) Embeddings of SBINet on Houston2013 dataset. (e) Embeddings of SBINet on PaviaU dataset. (f) Embeddings of SBINet on YC dataset. (g) Embeddings of HSDBIN on Houston2013 dataset. (h) Embeddings of HSDBIN on PaviaU dataset. (i) Embeddings of HSDBIN on YC dataset.

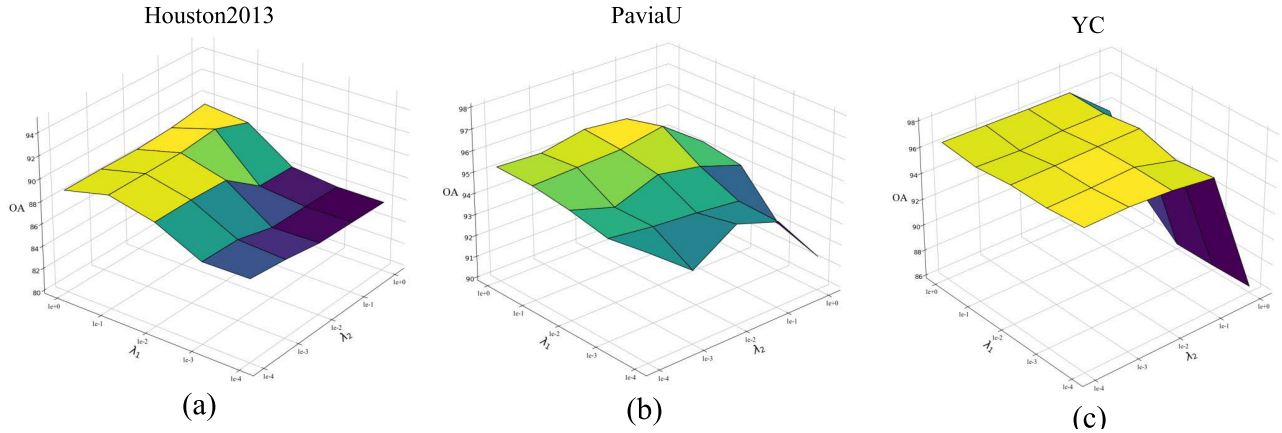


Fig. 11. OAs of HSDBIN with different values of λ_1 and λ_2 on three datasets. (a) Houston2013 dataset. (b) PaviaU dataset. (c) YC dataset.

we use the spherical coordinate to represent the distribution of features. It can be evidently found that the distribution of original hyperspectral data is seriously overlap (i.e., features coming from different categories are entangled with each

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other). Furthermore, sample points from the same class may exhibit loose distribution in high-dimensional space (i.e., there is no compact intraclass distance among the same class). In comparison, the features in the latent space of SBINet have larger interclass distances and more compact intraclass families, making the learned representations prone to be separated and recognized. This suggests that the SBINet can effectively extract task-discriminative spatial–spectral features.

However, examining the mean vectors of different classes reveals that both the raw data and the outputs from SBINet tend to be biased and concentrated in certain regions of the metric space, resulting in complex decision boundaries between some clusters (increasing the risk of overfitting). As expected, with the addition of HSD, the mean vectors of different classes are uniformly distributed throughout the entire metric space, significantly enhancing the utilization of the metric space. Therefore, the visualization results suggest that our proposed HSD not only preserves compact intraclass separability learned by the classifier but also significantly improves interclass distances.

E. Parameter Analysis

There are mainly two parameters λ_1 and λ_2 that need to be discussed. They are responsible for cooperatively optimizing the hyperspherical prototype learning and structural-aware distillation, respectively. Larger values indicate a greater need to attract more optimization energy toward the network. However, setting these parameters too large or too small will lead to overfitting or underfitting of certain tasks, then making a difference in final performance.

To explore the sensitivity of HSDBIN to these two hyperparameters, we selected two parameter values from the range $\{1e+0, 1e-1, 1e-2, 1e-3, 1e-4\}$ and examined their joint performance. Fig. 11 presents the classification performance of HSDBIN with different λ_1 and λ_2 on three datasets, and OA is chosen as the primary metric. The results from the figure clearly yield that as the values of λ_1 and λ_2 decrease, the classification performance of HSDBIN significantly declines. This is because when the values of these two parameters are too low, both the hyperspherical prototype learning and structural-aware distillation tasks are not adequately optimized. In this case, the capability of HSDBIN essentially degrades to role only performing $\mathcal{L}_{\text{Cls}}$ without leveraging the benefits of multitask learning and structural-aware optimization. In addition, when the values of these parameters are not too low (e.g., $1e+0$, $1e-1$, $1e-2$), the model's performance remains relatively consistent, indicating that our method is not highly sensitive to these parameters within a certain range.

IV. Conclusion

In this article, a novel HSDBIN is proposed. HSDBIN uses a dual-branch design that combined the 1-D CNN and transformer to fully learn the detailed spectral correlations and model global spatial relationship in parallel. To capture the complementary clues between spectral and spatial pipelines, a bidirectional interaction across branches is designed to interact and aggregate the independent information between two parallel branches. Finally, an HSD is proposed to enhance the utilization of metric space and preserve the highly structured relationship of output logits. Extensive experiments and analysis suggest that the proposed HSDBIN achieves superior performance on three benchmark datasets. Moreover, the results of ablation experiments verify the effectiveness of the proposed modules. The HSDBINs are based on the closed-set samples; in the future, we will apply this technique to the universal HSIC, such as partial set or more extreme open set.

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